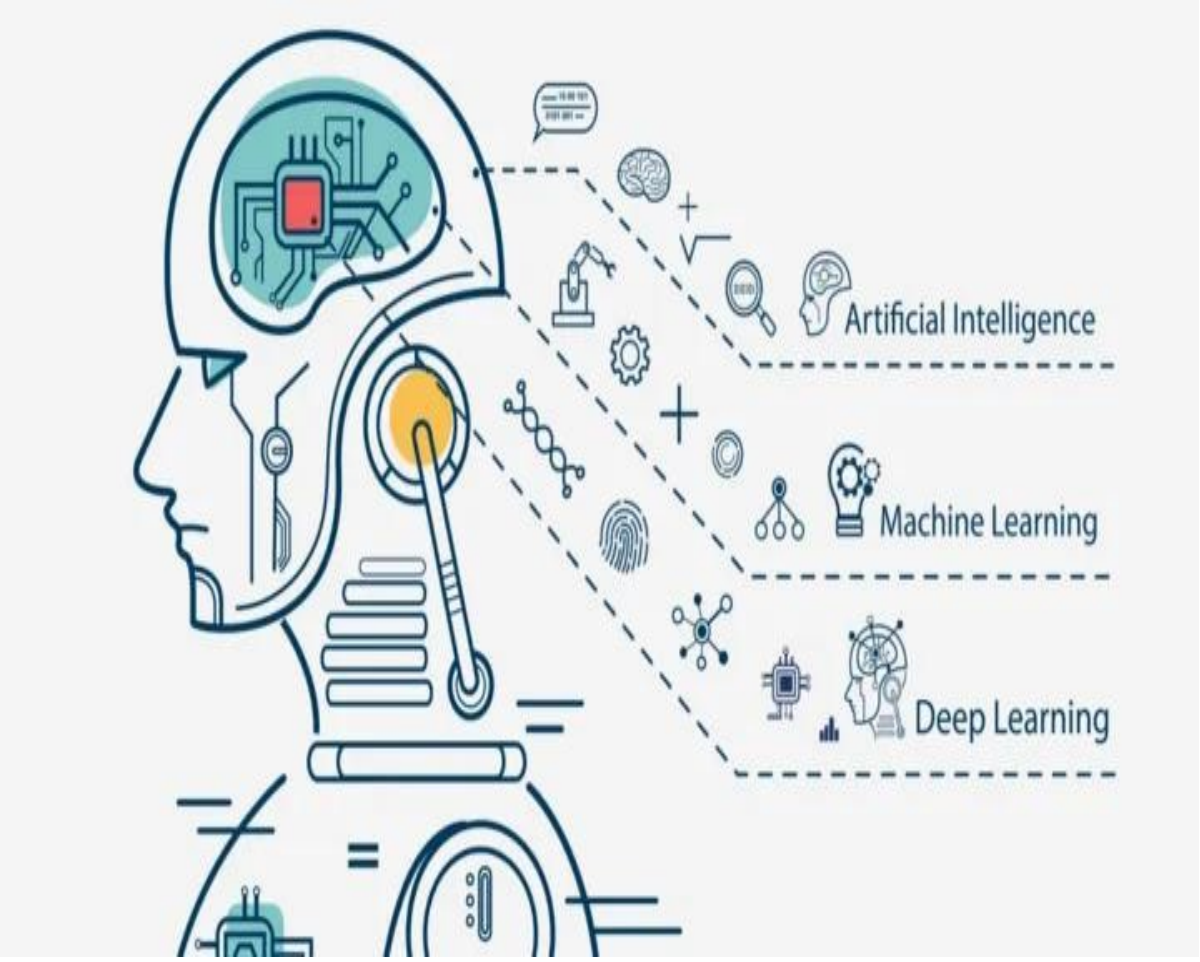
ABSTRACT

Real estate is the least transparent industry in our ecosystem. Housing prices keep changing in and day out and sometimes are hyped rather than being based on valuation. Predicting housing prices with real factors is the main crux of our research project. Here we aim to make our evaluations based on every basic parameter that is considered
while determining the price. We use various regression techniques in this pathway, and our results are not sole determination of one technique rather it is the weighted mean of various techniques to give most accurate results. The results proved that this approach yields minimum error and maximum accuracy than individual algorithms applied. We also propose to use real-time neighbourhood details to get exact real-world valuations. Since housing price is strongly correlated to other factors such as location, area, population, it requires other
information apart from "HOUSE PRICE PREDICTION" to predict individual housing price. Data mining is now commonly applied in the real estate market. Data mining's ability to extract relevant knowledge from raw data makes it very useful to predict house prices, key housing attributes, and many more. Research as stated that the fluctuation in house prices are often a concern for house owner and the real estate market. A survey of literature is carried out to analyze the relevant attributes and the most efficient models to forecast the house prices. The real estate market is a standout amongst the most focused regarding pricing and keeps fluctuating.

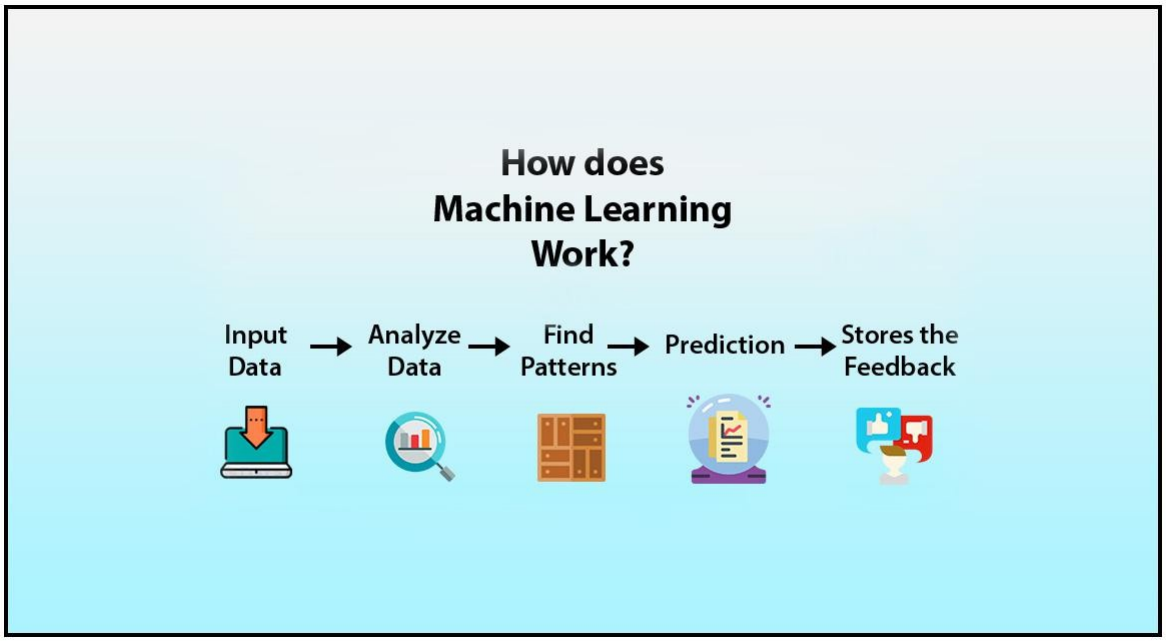
INTRODUCTION:

Machine learning is a subfield of artificial intelligence(Al). The goal of machine learning is to understand the structure of the data and fit that data into models that can be understood and utilized by people. Although machine leaming is a field within computer science it differs form traditional computational approaches. In traditional computing, algorithms are set of explicitly programmed instructions used by computers to calculate or problems solve. Machine learning algorithms instead allow for computers to train on data inputs and use statistical analysis in order to outputs values that fall within a specific range. Because of this, machine learning facilitates computers in building models from sample data in order to automate decision making processes based on data inputs.



How does Machine Learning work?

Machine learning is a form of artificial intelligence (AI) that teaches computers to think in a similar way to how humans do learn and
improving upon past experiences. It works by exploring data, identifying patterns, and involves minimal human intervention.
Almost any mask that can be completed with a daa-defined pattern or set of rules can be automated with machine learning. This allows companies to transform processes that were previously only possible for humans to perform think responding to customer service calls, bookkeeping and reviewing resumes.



Machine Learning Techniques:

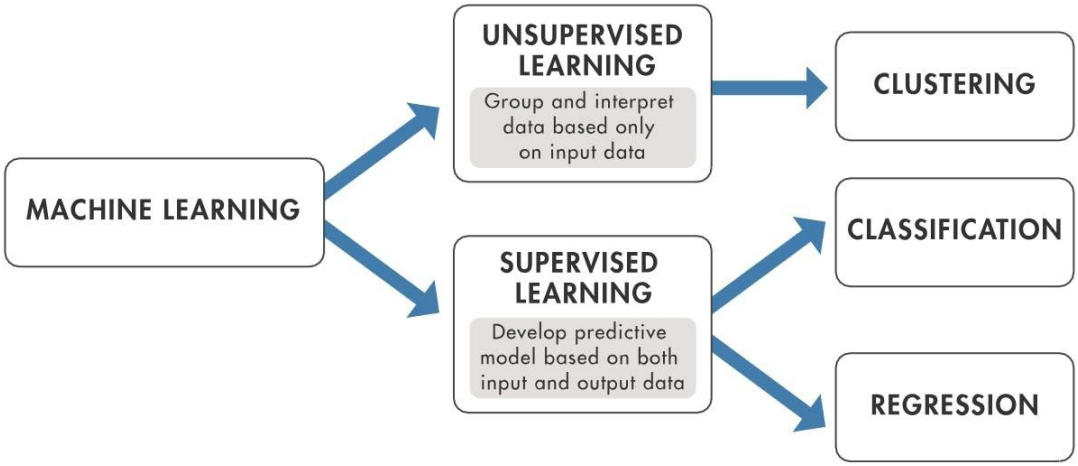
Machine learning uses two types of techniques:

Supervised learning:

which trains a model on known input and output data so that it can predict future outputs? It allows you to collect data or produce a data output from aprevious ML deployment. Supervised learning is exciting because it works in much the same way humans actually learn.

Unsupervised learning:

which finds hidden patterns or intrinsic structures in input data? Helps you find all kinds of unknown in data. In unsupervised learning. The algorithm tries to learn some inherent structure to the data with only unlabled examples. Two common unsupervised learning tasks are clustering and dimensionality reduction.



What is Learning? Rats learning to avoid poisonous Baits: Rats normally stumble upon food items by its look and smell and start eating in small amounts and later depending on food and physiological effect the feeding of food goes on. If the rat notices the illness of the food, the rat will not touch that food. Similarly the machine learning mechanism plays a vital role same as animal usage of past experience for acquiring and expertise in detecting the food safety. By mistake if the knowledge with the food is negatively labelled, the prediction the animal will also will be negatively affected and encounted in the future. With the inspiration of the previous example of successful learning we demonstrate a typical machine learning algorithm. Likewise we would want to program a machine that learns how to filter spam e-mails. A credulous results might be apparently comparable of the lifestyle of rats, how to keep away from poisonous baits. The machine will basically remember the past e-mails that named similarly as spam .

Objective:-

Machine learning plays a major role from past years in image detection, spam reorganization, normal speech command, product recommendation and medical diagnosis. Present machine learning algorithms helps us in enhancing security alerts, ensuring public safety and improve medical enhancements. Machine learning system also provides better customer service and safer automobile systems. In the present paper we discuss about the prediction of future housing prices that is generated by machine learning algorithm. For the selection of prediction methods we compare and explore various prediction methods. We utilize lasso regression as our model because of its adaptable and probabilistic methodology on model selection. Our result exhibit that our approach of the issue need to be successful, and has the ability to process predictions that would be comparative with other house cost prediction models. Move over on other hand housing value indices, the advancements of a housing cost prediction that tend to the advancements of real estate policies schemes.

SCOPE:

There are quite a few things that can be polished or add in the future scope. Though, we were able to identify most of the residential areas. There may be some more places that housing complexes or multi-store apartments which are located in commercial areas. Such apartments were not included in this paper and can be counted in future to give a more accurate result. With more and more demand for housing in metropolitan cities, there is a definite increase in the number private builders that provide real estate with additional amenities to attract more customers. There are several other models available that can be implemented for prediction. Data given as input to such model should be compatible with the tool used and the operators involved in the process. Also, more numbers of data sets can be used to increase.

HOUSE PREDICTION:

House Prediction using Machine Learning refers to using algorithms to estimate the price or value of a house based on various features. Here's a clear and simple explanatio

What is House Prediction?

It is a regression problem in machine learning where the goal is to predict house prices using input features like:

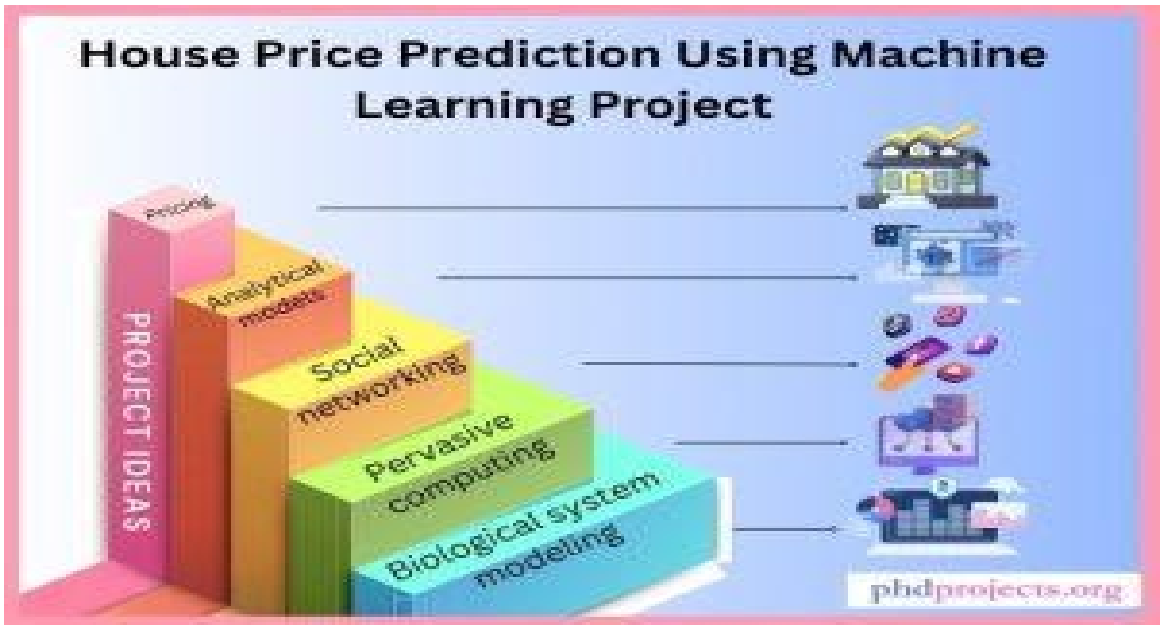
- Location
- Size (in square feet)
- Number of bedrooms/bathrooms
- Age of the house
- Nearby facilities (schools, transport)
- House condition, etc.

Example Use Case:

A real estate company wants to predict how much a house will sell for. They collect data on many houses and train a machine leaming model to predict prices for new listings.

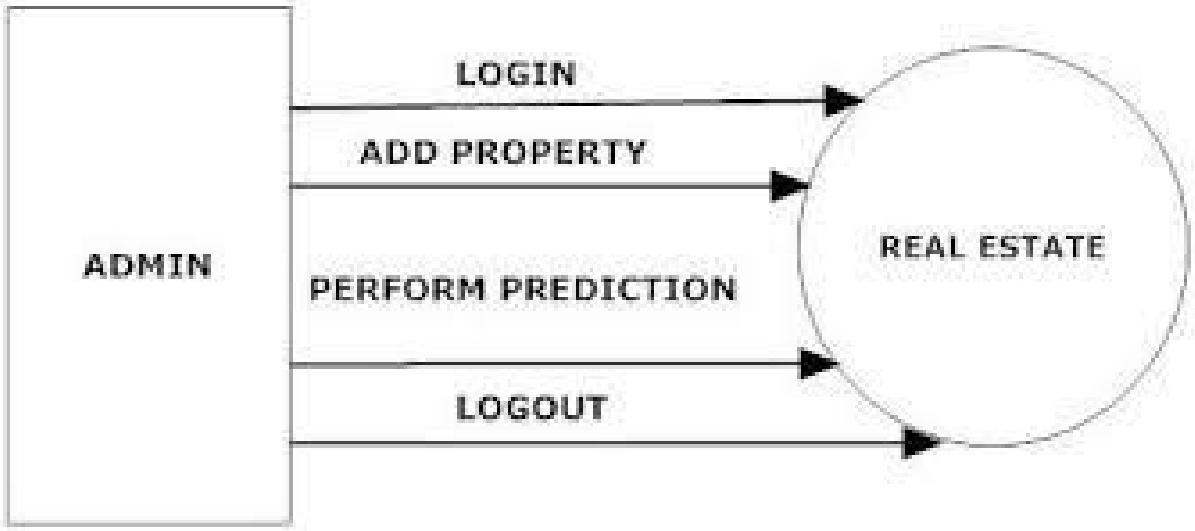
Common Machine Learning Algorithms Used:

1. Linear Regression - Simple model that finds a straight-line relationship.
2. Decision Tree Regressor - Splits data into tree-like structures based on features.
3. Random Forest - Many decision trees combined for better accuracy.
4. XG Boost - Advanced boosting model used in competitions.

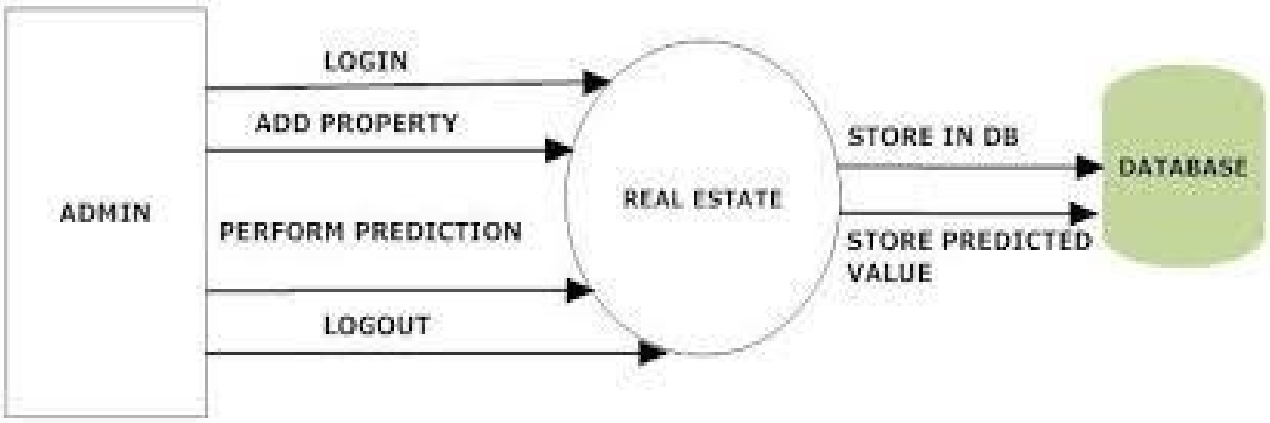


METHODOLOGY:

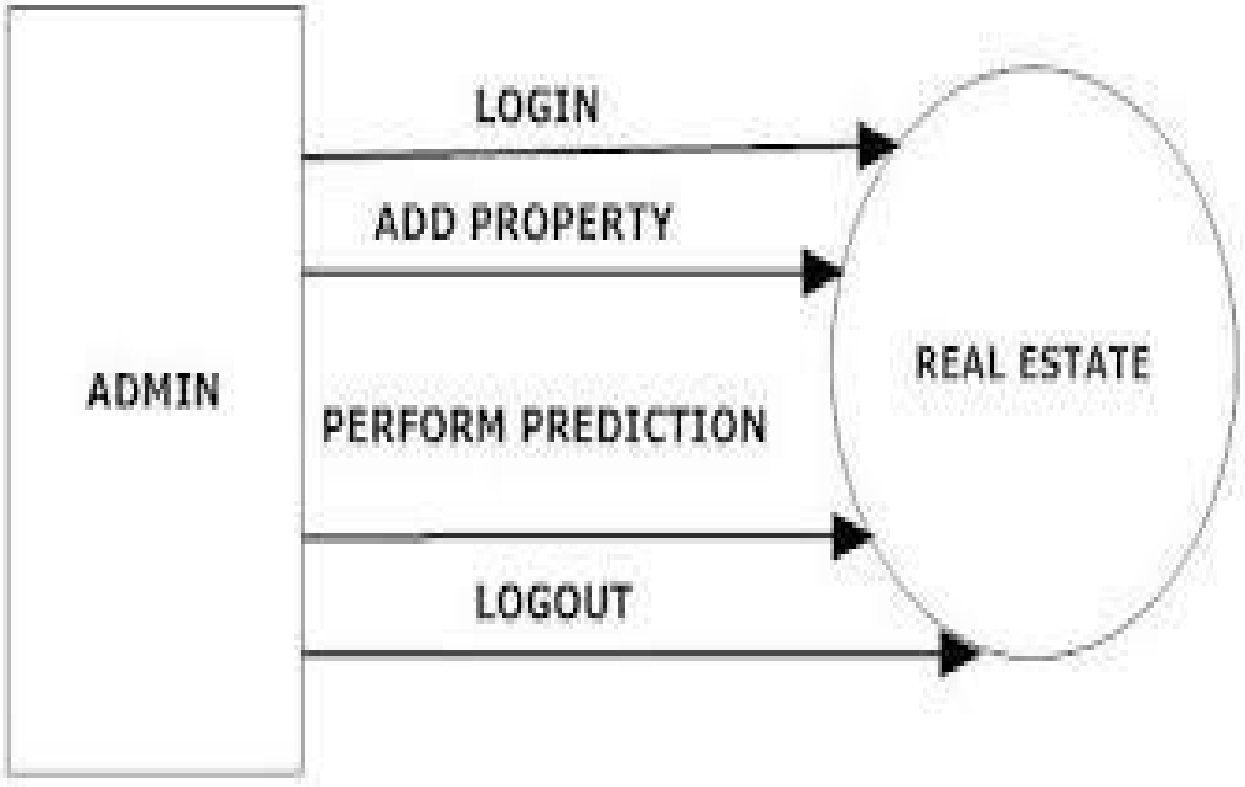
DFD DIAGRAM



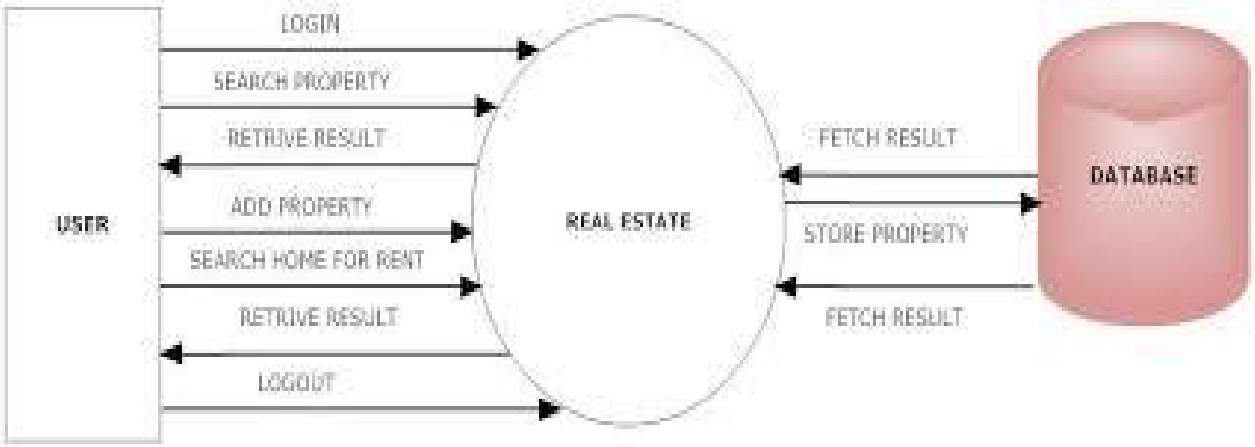
DFD admin level 1: Admin have to first log in into the website. When he enters new property then that info will be stored into the database so that it will be easy to retrieve. When the predictions are made, that data also stored into the database.



DFD user level 0:- After user is successfully logged in, he can search properties according to his preferences. He can add his/her shortlist properties. He can also search for respective rent for any provided property.



data from DFD user level 1: User can log in his account. We he search some property, system retrieves data from database and results are shown to user. When add some property it is stored by system in database. He can also search for rent of Houses. System will retrieve the requested the database and results will be shown to user.



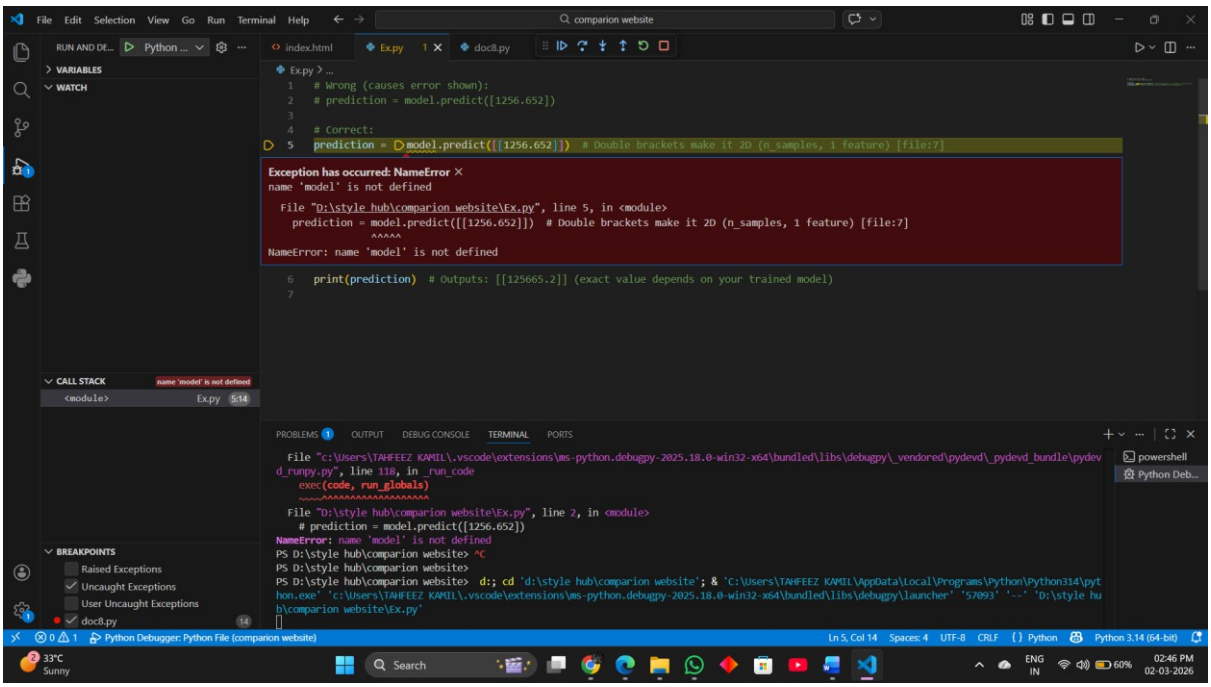
ADVANTAGES:

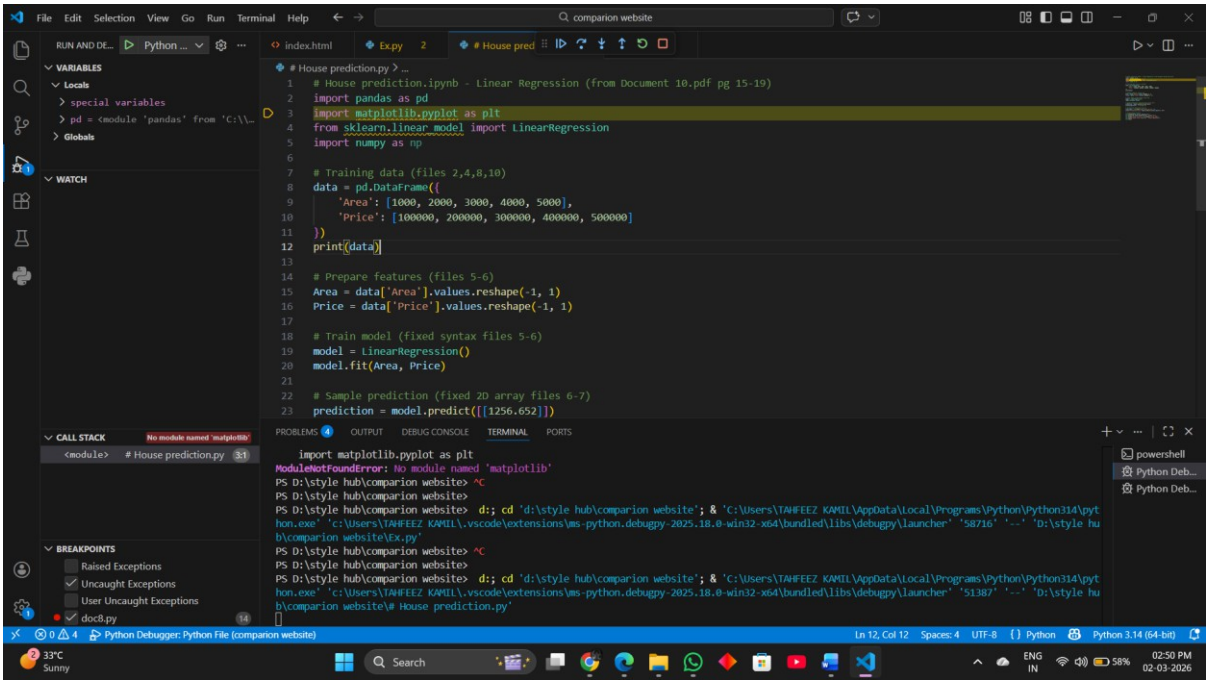
- House price prediction can help the developer determine the selling price of a house and can help the customer to arrange the right time to purchase a house
- There are three factors that influence the price of a house which include physical conditions, concepts and location.
- We would be able to better predict the prices.

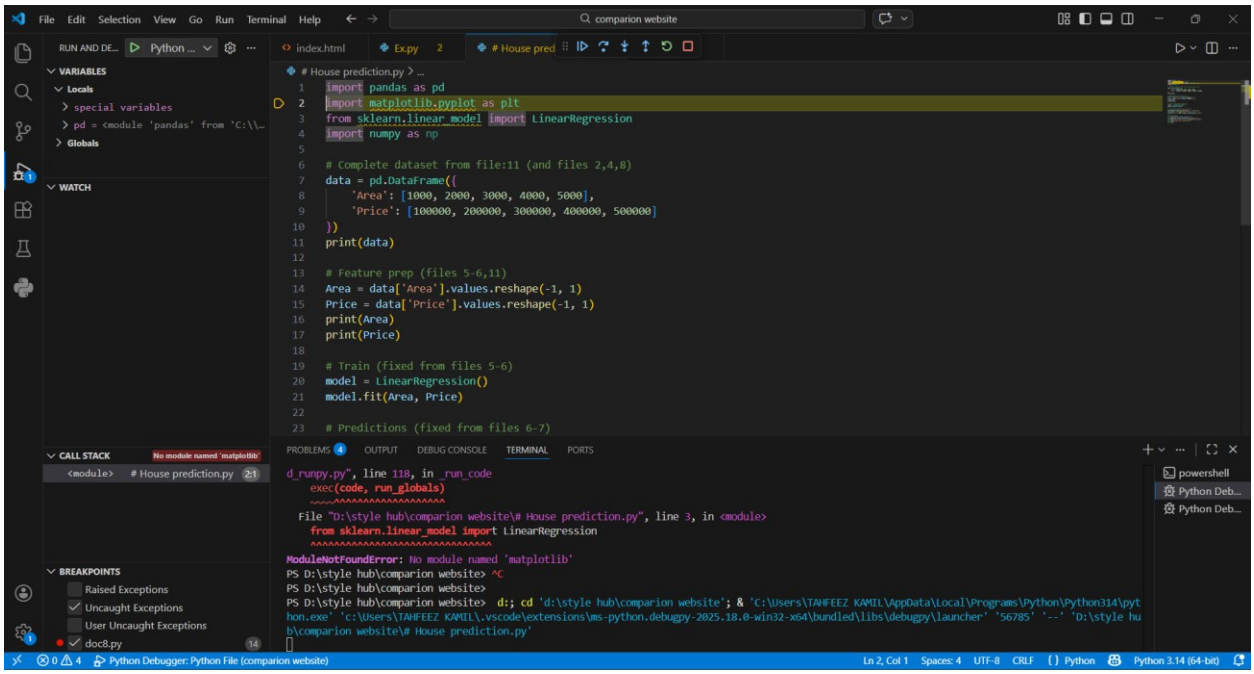
DISADVANTAGES:

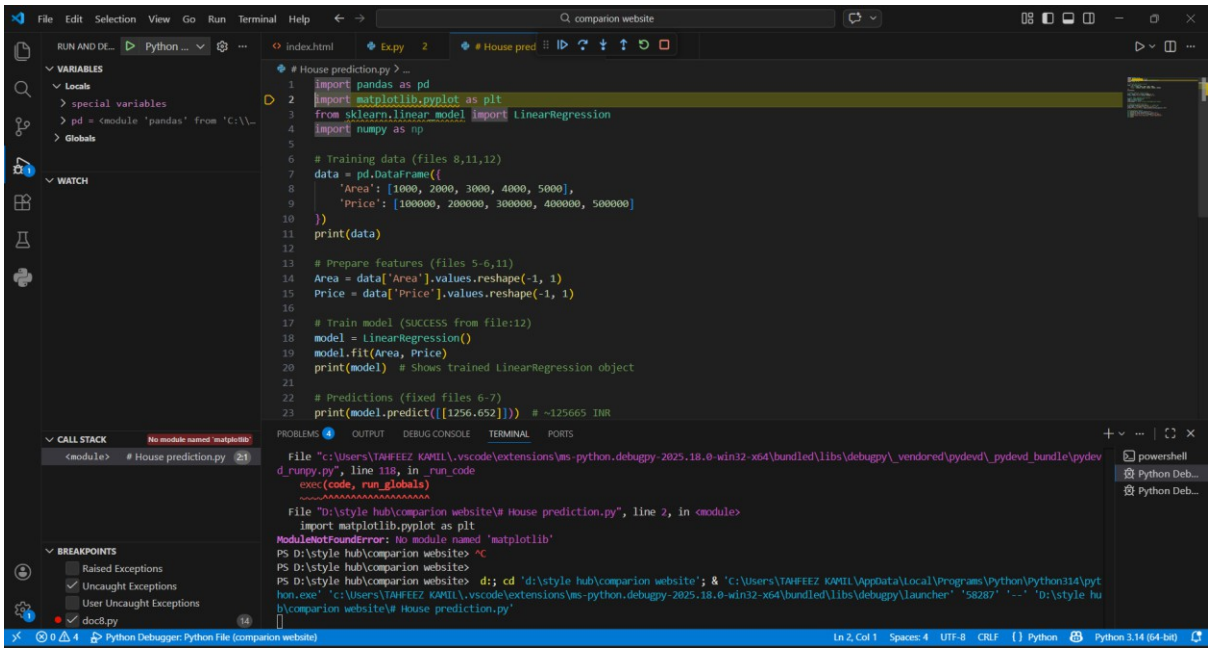
- The neural network required time for big neural networks.
- It fails in cases where mean and covariance are not enough to define datasets
- We need to choose an optimal kernel for SVM and this task is difficult.

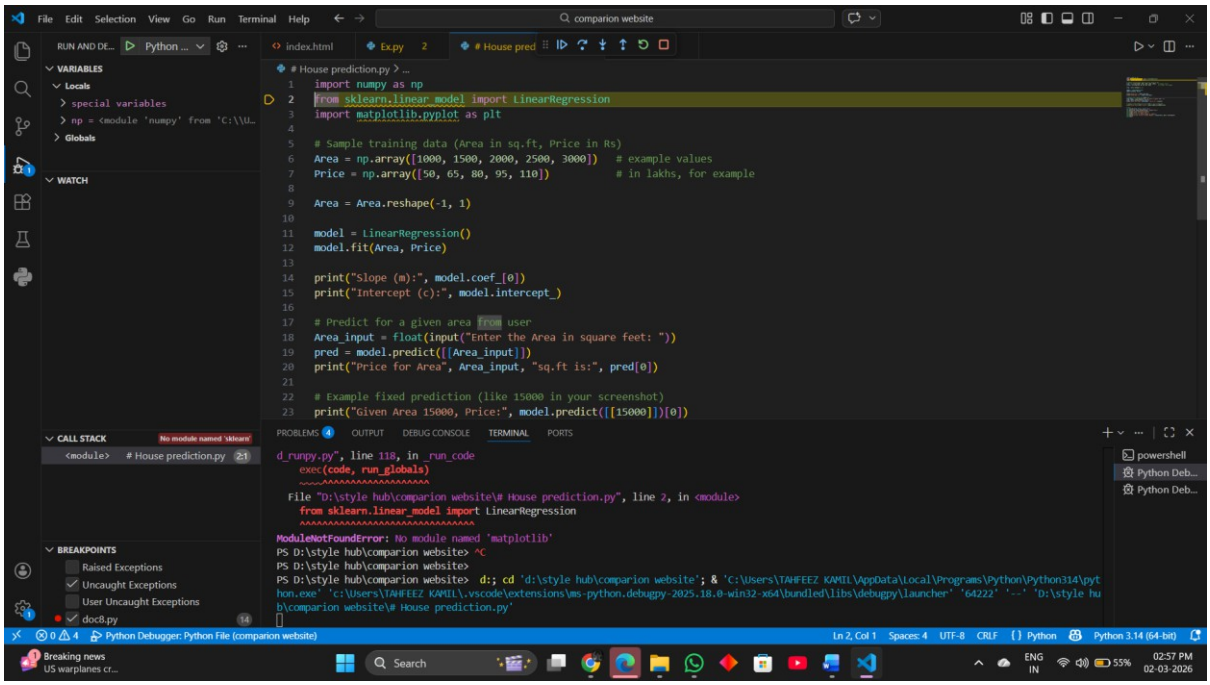
PROGRAM:

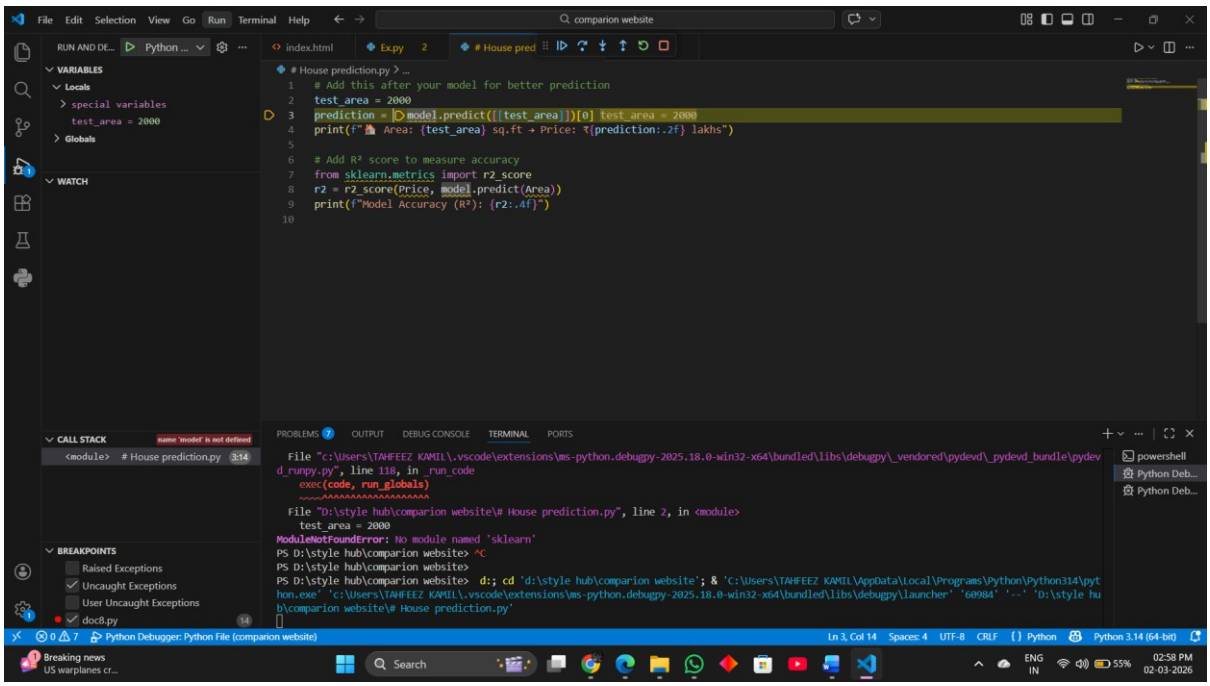

Exception has occurred: NameError
name 'model' is not defined
File "D:\style hub\comparion website\Ex.py", line 5, in <module>
prediction = model.predict([[1256.652]]) # Double brackets make it 2D (n_samples, 1 feature) [file:7]
NameError: name 'model' is not defined


ModuleNotFoundError: No module named 'matplotlib'


File Edit Selection View Go Run Terminal Help
comparion website
RUN AND DE... Python ...
VARIABLES
Locals
special variables
pd = <module 'pandas' from 'C:\\...
Globals
WATCH
CALL STACK
No module named 'matplotlib'
<module> # House prediction.py 2:1
BREAKPOINTS
Raised Exceptions
Uncaught Exceptions
User Uncaught Exceptions
doc8.py 14
index.html
Ex.py 2
House pred
House prediction.py > ...
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from sklearn.linear_model import LinearRegression
4 import numpy as np
5
6 # Complete dataset from file:11 (and files 2,4,8)
7 data = pd.DataFrame({
8 'Area': [1000, 2000, 3000, 4000, 5000],
9 'Price': [100000, 200000, 300000, 400000, 500000]
10 })
11 print(data)
12
13 # Feature prep (files 5-6,11)
14 Area = data['Area'].values.reshape(-1, 1)
15 Price = data['Price'].values.reshape(-1, 1)
16 print(Area)
17 print(Price)
18
19 # Train (fixed from files 5-6)
20 model = LinearRegression()
21 model.fit(Area, Price)
22
23 # Predictions (fixed from files 6-7)
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS
d_runpy.py", line 118, in _run_code
exec(code, run_globals)
File "D:\style hub\comparion website\# House prediction.py", line 3, in <module>
from sklearn.linear_model import LinearRegression
ModuleNotFoundError: No module named 'matplotlib'
PS D:\style hub\comparion website> ^C
PS D:\style hub\comparion website>
PS D:\style hub\comparion website> d:; cd 'd:\style hub\comparion website'; & 'C:\Users\TAHFEEZ KAMIL\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\TAHFEEZ KAMIL\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '56785' '--' 'D:\style hub\comparion website\# House prediction.py'
powershell
Python Deb...
Python Deb...
Python Debugger: Python File (comparion website)
Ln 2, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14 (64-bit)

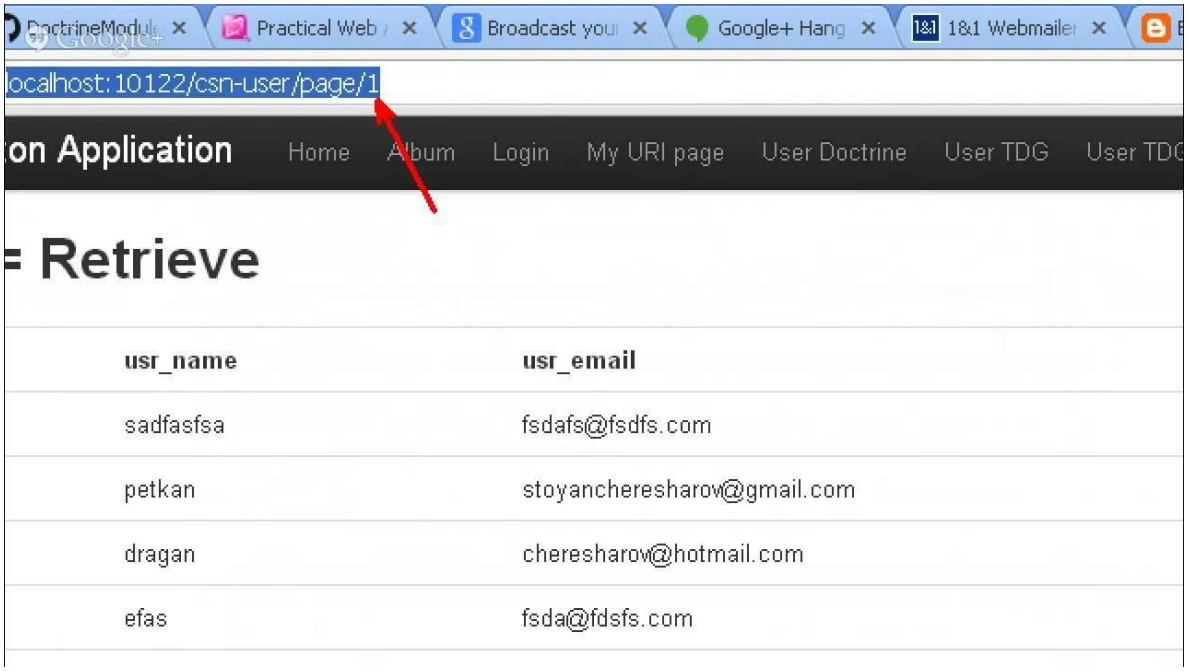

File Edit Selection View Go Run Terminal Help
comparion website
RUN AND DE... Python ...
VARIABLES
Locals
special variables
pd = <module 'pandas' from 'C:\\...
Globals
WATCH
CALL STACK
No module named 'matplotlib'
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BREAKPOINTS
Raised Exceptions
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User Uncaught Exceptions
doc8.py 14
index.html
Ex.py 2
House pred
House prediction.py > ...
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from sklearn.linear_model import LinearRegression
4 import numpy as np
5
6 # Training data (files 8,11,12)
7 data = pd.DataFrame({
8 'Area': [1000, 2000, 3000, 4000, 5000],
9 'Price': [100000, 200000, 300000, 400000, 500000]
10 })
11 print(data)
12
13 # Prepare features (files 5-6,11)
14 Area = data['Area'].values.reshape(-1, 1)
15 Price = data['Price'].values.reshape(-1, 1)
16
17 # Train model (SUCCESS from file:12)
18 model = LinearRegression()
19 model.fit(Area, Price)
20 print(model) # Shows trained LinearRegression object
21
22 # Predictions (fixed files 6-7)
23 print(model.predict([[1256.652]])) # ~125665 INR
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS
File "c:\Users\TAHFEEZ KAMIL\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy_vendored\pydevd_pydevd_bundle\pydevd_runpy.py", line 118, in _run_code
exec(code, run_globals)
File "D:\style hub\comparion website\# House prediction.py", line 2, in <module>
import matplotlib.pyplot as plt
ModuleNotFoundError: No module named 'matplotlib'
PS D:\style hub\comparion website> ^C
PS D:\style hub\comparion website>
PS D:\style hub\comparion website> d:; cd 'd:\style hub\comparion website'; & 'C:\Users\TAHFEEZ KAMIL\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\TAHFEEZ KAMIL\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58287' '--' 'D:\style hub\comparion website\# House prediction.py'
powershell
Python Deb...
Python Deb...
Python Debugger: Python File (comparion website)
Ln 2, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
comparion website
index.html
Ex.py 2
House pred
House prediction.py > ...
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
Sample training data (Area in sq.ft, Price in Rs)
Area = np.array([1000, 1500, 2000, 2500, 3000]) # example values
Price = np.array([50, 65, 80, 95, 110]) # in lakhs, for example
Area = Area.reshape(-1, 1)
model = LinearRegression()
model.fit(Area, Price)
print("Slope (m):", model.coef_[0])
print("Intercept (c):", model.intercept_)
Predict for a given area from user
Area_input = float(input("Enter the Area in square feet: "))
pred = model.predict([[Area_input]])
print("Price for Area", Area_input, "sq.ft is:", pred[0])
Example fixed prediction (like 15000 in your screenshot)
print("Given Area 15000, Price:", model.predict([[15000]])[0])
PROBLEMS 4
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
ModuleNotFoundError: No module named 'matplotlib'
PS D:\style hub\comparion website> ^C
PS D:\style hub\comparion website>
Python Debugger: Python File (comparion website)
Ln 2, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.14 (64-bit)
Search
02:57 PM
02-03-2026


File Edit Selection View Go Run Terminal Help
comparion website
House prediction.py > ...
Add this after your model for better prediction
test_area = 2000
prediction = model.predict([[test_area]])[0] test_area = 2000
print(f"🏠 Area: {test_area} sq.ft → Price: ₹{prediction:.2f} lakhs")
Add R² score to measure accuracy
from sklearn.metrics import r2_score
r2 = r2_score(Price, model.predict(Area))
print(f"Model Accuracy (R²): {r2:.4f}")
PROBLEMS 7
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
ModuleNotFoundError: No module named 'sklearn'
PS D:\style hub\comparion website> ^C
PS D:\style hub\comparion website>
Python Debugger: Python File (comparion website)
Ln 3, Col 14
Spaces: 4
UTF-8
CRLF
Python
Python 3.14 (64-bit)
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02-03-2026

SCREENSHORTS:





4.4. Screenshots regarding project

4.4.1 DASHBOARD





Conclusion:

In today real estate world, it has become tough to store such huge data and extract them for ones own requirement. Also, the extracted data should be useful. The system makes optimal use of the data mining algorithm. The system makes use of such data in the most efficient way. The data mining algorithm helps to fulfil customers by increasing the accuracy of estate choice and reducing the risk investing in an estate. A lot of features that could be added to make the system more widely acceptable. One of the major future scopes is adding estate database of more cities which will provide the user to explore more estates and reach an accurate decision. More factors like recession that affect the house prices shall be added. In-depth details of every property will be added to provide ample details of a desired estate. This will help the system to run on a larger level. There are quite a few things that can be polished or add in the future work. Though, we were able to identify most of the residential areas. There may be some more places that have housing complexes or multi-story apartments which are located in commercial areas. Suh apartments were not included in this paper and can be counted in future to give a more accurate result. With more and more demand for housing in metropolitan cities, there is a definite increase in the number private builders that provide real estate with additional amenities to attract more customers. There are several other models available that can be implemented for prediction. Data given as input to such model should be compatible with the tool used and the operators involved in the process. Also, more number of data sets can be used to increase the accuracy of the model.